Chapter 17 exercises • Distributed systems

OS17-E01 • Speedup and fault domains

A 120 s job has 30 s serial work, 90 s evenly divided over six nodes, and 6 s coordination. In a separate model, three interchangeable replicas have independent probabilities 0.02 of being unavailable at the same fixed read instant. Networking and metadata are available, and any available replica returns valid data.

1. Compute T6, speedup, and the infinite-node bound without coordination.
2. Find read unavailability at that instant. Why may shared-rack failures invalidate multiplication?

OS17-E02 • Moving data or computation

Data is 512 MiB, result 2 MiB, throughput 64 MiB/s, RTT 10 ms, and remote compute 200 ms. Charge exactly one RTT per option; transfer and computation do not overlap. Ignore request size, queues, other costs, and local compute time.

1. Compare full-data transfer time with remote computation plus result transfer.
2. Is sending code migration of a running process?

Lin Cing https://github.com/xinhi6716/computer-science-self-study

OS17-E03 • BDP and windows

Link200 Mbit/s, RTT50 ms, outstanding limit128 KiB, no loss/other bottleneck.

1. Find BDP and window throughput bound.
2. Does upgrading the link to1 Gbit/s multiply that window bound by five?

OS17-E04 • DNS cache and identity

At t=0 a resolver caches store.lab.test A=198.51.100.20 with TTL 300 s. The authority changes it to .30 at t=100 and makes no later change. The resolver obeys TTL, does not serve stale or apply local overrides, and its query after expiry succeeds.

1. What is returned at t=150 and after expiry at t=300? Must every label have a server?
2. Are domain, IP, and port equivalent to user, PID, or permanent host identity?

OS17-E05 • Routes and packet pipelines

An undirected graph has fixed nonnegative costs A–C=2, C–B=2, A–D=1, D–B=5. Separately, two initially idle 1 Mbit/s links use FIFO complete-packet store-and-forward. A source sends four 1000 B packets consecutively; ignore headers, propagation, other queues, and processing costs.

1. Trace tentative B distances and the route after C–B fails.
2. Compare completion time with one 4000 B message.

OS17-E06 • IPv4, ARP, and a short frame

In two Ethernet LANs without NAT, A=192.0.2.10/24 with MAC A; router R has left 192.0.2.1/MAC R_L and right 198.51.100.1/MAC R_R; B=198.51.100.20/24 has MAC B. A routes via R's left interface; A and R start with empty ARP caches. IPv4 and TCP each have 20 B headers without options; application payload is 4 B. Use untagged Ethernet II at 100 Mbit/s, 8 B preamble/SFD and 12 byte-times IFG.

1. Whom does A ARP? Give both hops' source/destination MAC and IP.
2. Find padding, MAC frame length, and wire occupancy including preamble/SFD and IFG.

OS17-E07 • Indistinguishable failure observations

A has no reply 200 ms after a ping. Part 1 gives no network or scheduling delay bounds.

1. Give equal local A histories with different B liveness.
2. Separately, healthy B schedules heartbeats at fixed nominal times H=0.5 s apart; each send delay is in [0,0.1] s, one-way network delay in [0,0.05] s, all messages arrive, receive processing is immediate, and the receiver clock has no drift. Find the maximum adjacent arrival gap.

OS17-E08 • Fencing and retries

A holds epoch 42, and a correct coordinator may grant B epoch 43, issuing unique monotonically increasing epochs. Counter starts at 10; the reply to increment(id17) may be lost. Storage orders epoch installation and each write's validation and commit at one atomic linearization point. The request ID is unique in the client/service namespace, its record survives every possible retry, and the record and counter share one durable transaction.

1. Why is timeout-based self-election unsafe? State resource-side fencing conditions.
2. How does retry preserve 11 instead of incrementing to 12? What crash windows occur if the durable updates are separate?

OS17-E09 • DFS naming and cache cost

Path→FID(vol9,file2,g4), vol9→serverA. One hundred reads execute serially without overlap: 20 misses cost 6 ms each, 80 hits 0.2 ms each. Each miss transfers exactly 4 KiB, without retries or other traffic. Optional server validation adds exactly 1 ms to each read with no overlap.

1. Which map changes on movement to serverB? What must happen when file2 is reused?
2. Compute total delay and payload traffic, then the delay with validation.

OS17-E10 • Consistency and partitions

C1 caches v0 at t=0; C2 completes write v1 at t=2; C1's next periodic validation is t=8. A new read begins at t=3; there are no other writes and C1's cache remains valid under this periodic model.

1. Can the read return cached v0? Does this model guarantee immediate visibility of a completed write?
2. A separate callback/lease model requires new reads after a completed write to exclude v0. During a partition, what recall acknowledgment or lease, clock, and client conditions are needed before write completion?

OS17-E11 • NFSv3 verifier changes

An NFSv3 client receives UNSTABLE and verifier X for WRITE D; before COMMIT the server restarts, loses unstable D, and uses verifier Y. A WRITE can be partial and the client retains all bytes not yet known stable.

1. What follows when COMMIT returns Y≠X? Under what conditions can successful COMMIT or stable WRITE release the retransmission obligation for covered bytes?
2. Does stateless mean no cache or every WRITE synchronously persists?

OS17-E12 • Historical HDFS and failover capacity

Two independent models. HDFS uses Hadoop 1.2.1, file 150 MiB, blocks 64 MiB, replication factor 3, at least three eligible DataNodes, and three completed replicas of every block. Separately, two request servers each serve 100 requests/s of equal-cost work; offered load is 150/s and can be balanced. These servers do not limit the number of HDFS DataNodes.

1. Give block payloads, total completed replica payload, and NameNode/DataNode paths.
2. Can one surviving request server sustain the same load?

References and attribution

Lin Cing, handwritten operating systems notes. Silberschatz et al., Operating System Concepts, 9th ed., Ch. 17. RFC 1034, RFC 826, RFC 1813. Hadoop 1.2.1 HDFS. OpenAFS.

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